
ADAM R. HUFF

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SUMMARY:

Protein biochemist with 8+ years in academic and pharmaceutical R&D. Specializes in recombinant protein production workflows comprising construct design, expression, purification, and biophysical characterization across a diverse range of complex drug discovery targets and host systems. Recent M.S. in Data Science graduate focusing on data management/architecture, statistical modeling, and applied machine learning. Co-author on publications in Nature, PNAS, and Nature Communications. Seeking to apply protein science expertise and computational modeling methods to problems in protein engineering and drug discovery.

EXPERIENCE:**GROW Rotation (In Silico Discovery) – Johnson & Johnson Innovative Medicine, Spring House, PA**

Sep 2026 – Present

- Building data pipelines and machine learning tools for protein engineering, including predictive modeling of expression outcomes and protein language model-based construct design.

Associate Scientist II – Johnson & Johnson Innovative Medicine, Spring House, PA

Apr 2019 – Present

- Co-developed a high-throughput expression screening pipeline integrating automated cell culture and expression in mammalian (HEK293/CHO) and insect (BEVS) cells, small-scale purification, and QC, cutting screening lead time by ~50%.
- Designed dozens of expression construct variants for challenging drug discovery targets, optimizing domain boundaries and tagging strategies; developed protein standards now used group-wide for quantification.
- Expressed, purified, and characterized diverse modalities (non-antibody proteins, mAbs, Fabs, nanobodies, Fc-fusions) across 350+ screens enabling assay development, structural biology, and hit-finding efforts.
- Built a production Python ETL pipeline (SQLAlchemy, oracledb, pandas) for DNA inventory management integrating an AWS-hosted Oracle database, REST API, and local sources, reducing hands-on time from hours to minutes.
- Contributed as scientific stakeholder in gene-to-protein database migration; cleaned, transformed, and bulk-generated thousands of entities, establishing complete gene-to-protein sample lineages.
- Led validation and integration of capillary electrophoresis (ProteinSimple Wes/Jess/Leo, Revvity LabChip GXII Touch) and analytical FSEC platforms; scripted a CyBio Felix liquid handler to automate CE sample preparation.
- Collaborated cross-functionally with scientists, automation engineers, and external CROs; trained contract scientists on expression and characterization workflows.

Scientist I – Kelly OCG servicing Janssen R&D, Spring House, PA

Mar 2018 – Apr 2019

- Conducted recombinant protein expression screening and scale-up across bacterial (E. coli), insect (BEVS/Sf9), and mammalian (HEK293/CHO) systems in support of drug discovery pipeline deliverables.
- Assessed expression and protein integrity via SDS-PAGE and Western blot across multiple constructs and conditions to support candidate selection and scale-up decisions.

Research Assistant – Department of Chemistry, Haverford College, Haverford, PA

Jan 2017 – Mar 2018

- Conducted NIH-funded research on acyl carrier protein fatty acid chain sequestration; mentored undergraduate researchers.

SKILLS:

Languages: Python, SQL, R, Bash

Libraries: pandas, NumPy, scikit-learn, TensorFlow/Keras, RDKit, SQLAlchemy, oracledb, psycopg2, requests, Flask, Matplotlib, Seaborn, tidyverse/ggplot2

Protein Modeling: AlphaFold/ColabFold, ESM/ESMFold, ProteinMPNN, RFdiffusion, PyMOL (pLDDT/PAE model evaluation)

Methods: Regression, classification, clustering, neural networks, dimensionality reduction, feature engineering, model evaluation (cross-validation, AUC, RMSE, R^2), ETL pipelines, relational schema design, REST API integration

Tools and Platforms: Git, Docker, PostgreSQL, Oracle, Linux (Ubuntu), AWS, Tableau Cloud, Jupyter

Domain Expertise: Protein Biochemistry, Structural Biology, Drug Discovery, Experimental Design, High-Throughput Screening, Laboratory Automation, Analytical Instrumentation

COMPUTATIONAL PROJECTS:

Protein Biochemical Profiling Pipeline

URL: <https://github.com/adamhuffds/biofetch>

- Built a modular CLI protein profiling tool in Python that queries the UniProt REST API, parses nested JSON responses, and displays structured protein profiles including identity, function, disease associations, and sequence data in a rich formatted terminal interface
- Implemented an interactive shell using Python's cmd module with a clean separation between data retrieval and display layers, leveraging the rich library for styled tables and panels across multiple biological data sources

Novozymes Enzyme Stability Prediction

URL: https://github.com/adamhuffds/protein_thermostability_modeling

- Built an end-to-end regression pipeline predicting enzyme thermostability from amino acid composition features derived from the Novozymes Kaggle dataset, including data cleaning, feature engineering, and model evaluation.
- Compared multiple regression models (Random Forest, Linear) using RMSE and R^2 , achieving $R^2 \sim 0.63$ with Random Forest

D2 Receptor ChEMBL Ingestion Pipeline

URL: https://github.com/adamhuffds/d2_receptor_pipeline

- Built a multi-endpoint data ingestion pipeline querying the ChEMBL REST API for IC50 bioactivity data, compound structures, and Lipinski physicochemical descriptors across paginated API calls
- Designed and implemented a normalized PostgreSQL schema (compounds, assays, bioactivity) with idempotent upsert logic via SQLAlchemy 2.0
- Conducted exploratory analysis of 1,580 IC50 records documenting pIC50 distributions, descriptor correlations, and data quality flags informing downstream pre-processing decisions

EDUCATION:

M.S., Data Science (2026; 3.9 GPA) – Eastern University

B.S., Cell Biology/Biochemistry (2017; *cum laude*) – Bucknell University

A.A., Chemistry (2014; highest honors) – Harrisburg Area Community College

PUBLICATIONS:

Nature (2025) Lamprecht, D.A., ... Huff, A., et al. Targeting *de novo* purine biosynthesis for tuberculosis treatment. *Nature*, 644(8075), 214–220. <https://doi.org/10.1038/s41586-025-09177-7>

PNAS (2023) Thompson, A.A., ... Huff, A.R., et al. Identification of small-molecule protein–protein interaction inhibitors for NKG2D. *Proceedings of the National Academy of Sciences*, 120(18), e2216342120. <https://doi.org/10.1073/pnas.2216342120>

Nature Communications (2019) Epstein, S.C., Huff, A.R., et al. Tracking carrier protein motions with Raman spectroscopy. *Nature Communications*, 10, 2227. <https://doi.org/10.1038/s41467-019-10184-2>

AIChE Journal (2018) Rivas, M., ... Huff, A.R., et al. The effect of divalent cations on the thermostability of type II polyketide synthase acyl carrier proteins. *AIChE Journal*, 64(12), 4308–4318. <https://doi.org/10.1002/aic.16402>

PLOS Biology (2017) Cookmeyer, D.L., ... Huff, A.R., et al. Uncovering protein–protein interactions through a team-based undergraduate biochemistry course. *PLOS Biology*, 15(11), e2003145. <https://doi.org/10.1371/journal.pbio.2003145>

AWARDS AND HONORS:

DTMP Innovation Award (2023), American Society for Microbiology Undergraduate Research Fellowship (2016), Multiple academic research scholarships and honors